Francisco Zambrano PhD

- Providencia, Santiago, Chile
- Chilean-Italian

Skills

- **Programming (Python/R/Matlab)** — 10+ yrs.
- **ML/DL (Python/R)** — 5+ yrs.
- **MySQL / PostgreSQL** — 5+ yrs.
- **CI/CD** — 5+ yrs.
- **Claude Code (CLI) + GitHub** — 0.5+ yrs
- **Google Earth Engine** — 3 yrs,
- **Amazon Web Service (AWS|EC2)** — 3 yrs.
- **Dockers** — 3 yrs.
- **Reproducibility (Rmarkdown/Quarto)** — 6 yrs.
- **Prototyping (R-Shiny/Python-Streamlit)** — 5 yrs.

Python libraries

- NumPy, Pandas, Matplotlib
- GeoPandas, rasterio, xarray, rioxarray, fiona
- scikit-learn, PyTorch
- streamlit

R packages

- tidyverse
- terra,sf,tmap
- tidymodels
- shiny

Summary

Senior Data Scientist specializing in environmental, agricultural, and climate risk systems, with a focus on decision-making and scaling analytical models. Over 15 years of experience in remote sensing, spatial analysis, and applied data science. I have led ANID-funded R&D projects totaling over CLP 600 million, developing national data platforms (ODES-Chile, SatOri) for drought monitoring, water management, and irrigation optimization. I possess strong expertise in Python, R, GIS, and machine learning to transform complex data into operational and scalable solutions in AgTech and climate risk contexts. I am seeking roles in the industry where I can translate scientific knowledge into real and measurable impact.

Experience

Senior Geospatial Scientist & Project Lead — Feb 2018 - Aug 2025

Earth Observation Center
Hemera - Universidad Mayor

- Built and launched ODES-Chile (https://odes-chile.org) - national drought observatory & early-warning system processing ERA5-Land, MODIS, CHIRPS at country scale (1,000-10,000 monthly users including Ministry of Agriculture).
- Designed and deployed SatOri (https://s4tori.cl) - operational satellite irrigation optimization platform for cherry orchards using Sentinel-2 + meteorological data + ML to predict stem water potential and deliver irrigation recommendations.
- Developed and released national PM2.5 forecasting dashboard (2025) combining SINCA stations + satellite aerosol data + ML (https://frzambra.shinyapps.io/app_pm25).
- Secured and directed 600M CLP (US$650k) in competitive grants as Principal Investigator.
- Published lead-author papers in Remote Sensing of Environment, Agricultural Water Management, Earth's Future

Visiting Doctoral Researcher — Sep - Dec 2016

Faculty of Geo-Information Science and Earth Observation (ITC)
University of Twente, Enschede, The Netherlands

- Built agricultural productivity decline models using MODIS/CHIRPS time-series and spatial analytics; published in Remote Sensing of Environment (impact: policy-relevant for Chilean drought mitigation).

Visiting Doctoral Researcher — Jan-Jun 2016

CALMIT/NDMC
University of Nebraska, United States

- Evaluated satellite precipitation products for drought monitoring; results in Atmospheric Research, informing national agroclimate reports.

Latest Publications

1. **Zambrano, F.**, Herrea, A., Molina-Roco, M. Explainable Machine Learning for Wheat Biomass Integrating Sentinel-1/2, PlanetScope and In-Situ Weather Data. 2026. **Remote Sensing Applications: Society and Environment (F.I. 4.5).** (Under review). `https://doi.org/10.31223/X5KJ1K`
2. **Zambrano, F.**, Anton, V., Meza, F., Duran-llacer, I., Fernández, F., Venegas-González, A., Raab, N., Craven, D., 2025. From Drought to Aridification: Land-cover fingerprints of a drying Chile. **Earth's Future (F.I. 8.2).** `https://doi.org/10.1029/2025EF006744`

Spatial data

- MODIS
- ERA5/ERA5-Land
- CHIRPS
- Sentinel-1/2
- Landsat 7/8/9
- SoilGrid
- CMIP6

Languages

- English - Advanced (B2-C1)
- Spanish - Native

Education

03/2014 - 09/2017

Ph.D in Agricultural Engineering, Mention in Water Resources
Universidad de Concepción

03/2000 - 09/2007

Civil Engineering

Universidad de Concepción

Contact

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Researchgate

LinkedIn

Github

3. **Zambrano, F.**, Herrera, A., Olguín, M., Miranda, M., Garrido, J., & Almeida, A. M. (2025). Prediction of the daily spatial variation of stem water potential in cherry orchards using weather and Sentinel-2 data. **Agricultural Water Management (F.I. 6.5)**, 318, 109721. https://doi.org/10.1016/j.agwat.2025.109721

4. Duran-Llacer, I., Gómez-Escalonilla Canales, V., Aliaga-Alvarado, M., Arumí, J.L., **Zambrano, F.**, Rodríguez-López, L., Martínez-Retureta, R., Martínez-Santos, P., 2025. Approach to mapping groundwater-dependent ecosystems through machine learning in central Chile. **Groundwater for Sustainable Development (F.I. 5.6)** 31, 101526. https://doi.org/10.1016/j.gsd.2025.101526

5. Duran-Llacer, I., Salazar, A. A., Mondaca, P., Rodríguez-López, L., Martínez-Retureta, R., **Zambrano, F.**, Llanos, F., & Frappart, F. (2025). Influence of Avocado Plantations as Driver of Land Use and Land Cover Change in Chile's Aconcagua Basin. **Land (F.I. 3.2)**, 14(4), 750. https://doi.org/10.3390/land14040750

6. **Zambrano, F.**, 2023. Four decades of satellite data for agricultural drought monitoring throughout the growing season in Central Chile, in: Vijay P. Singh Deepak Jhajharia, R.M., Kumar, R. (Eds.), Integrated Drought Management, Two Volume Set. CRC Press, p. 28.

7. Fernández, F. J., Vásquez-Lavín, F., Ponce, R. D., Garreaud, R., Hernández, F., Link, O., **Zambrano, F.**, & Hanemann, M. (2023). The economics impacts of long-run droughts: Challenges, gaps, and way forward. **Journal of Environmental Management (F.I. 8.4)**, 344, 118726. https://doi.org/10.1016/j.jenvman.2023.118726